Test-Time Score Distillation: Mask-Free Semantic Style Editing with Reverse-Process Regularization

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Abstract

Training-free style transfer with a frozen text-to-image transformer conflates two useful outcomes: preserving a reference composition while changing its subject, and transferring only the reference medium to a new composition. We formulate them as endpoints of a single *style-locality* axis and control both with **Test-Time Score Distillation (TSD)**. TSD evaluates a same-state contrast between reference-conditioned and reference-free frozen velocities, converts it into a fixed-magnitude rms-normalized latent update, and interleaves that update with the model's ordinary reverse steps. This gives an empirical *regularization substitution*: in our single-sample test-time regime, the frozen reverse trajectory repeatedly re-anchors the edited latent, while an added learned fake-score term suppresses structured media rather than improving stability. The number of reverse steps then acts as a locality control: more steps preserve reference composition for mask-free semantic style editing; fewer steps dissolve composition for global style transfer. We evaluate style fidelity, subject fidelity, and composition transfer separately using CSD, CLIP-T, and layout leak. Across 40 references, TSD improves CSD from 0.267 to 0.719 in the composition-preserving setting and from 0.247 to 0.615 in the composition-dissolving setting, winning all 40 paired styles in both cases, with a modest CLIP-T decrease of about 0.02. A five-style, four-prompt extension yields 20/20 additional paired wins. TSD trains no weights, but remains an optimization-based test-time method.

Introduction

Style transfer on a frozen generator is treated as one task, but users want two. Given a content prompt and a reference image, a frozen text-to-image transformer can render the prompt's subject "in the style of" the reference. In practice two very different outcomes are desirable and are conflated under one name. In one, an artist wants the reference's *scene* preserved — its composition, its arrangement — with only the subject re-imagined in the reference's medium; this is **Semantic Style Editing**. In the other, the artist wants only the reference's *look* — its medium, texture, and palette — carried onto a freshly composed scene, with the reference's specific layout thrown away; this is **Global Style Transfer**. Conflating them yields methods that silently do one and fail the other.

These two outcomes are the endpoints of a single controllable axis, which we call style locality. Style locality asks *how much of the reference is transferred*: at the local end, everything is transferred (medium and composition), giving semantic editing; at the global end, only the transferable, position-free part (medium, palette, texture statistics) is transferred, giving global transfer. Framing the problem as one axis, rather than two disjoint tasks, is what lets a single operator serve both — and lets a user dial between them.

We drive this axis with one frozen-model operator, derived rather than engineered. Recent distribution-matching distillation (Yin et al. 2024a) separates its update into a *spear* (a classifier-free-guidance- augmentation term that sharpens toward the conditioning) and a *shield* (a distribution-matching term, requiring a separately trained fake-score network, that keeps samples on the data manifold) (Liu et al. 2025). We find that for *test-time, single-sample* generation over a multi-step frozen reverse process, the ordinary inference dynamics provide sufficient regularization in our experiments, while adding a fake-score term suppresses structured style. Removing that auxiliary network leaves a pure frozen-teacher update, which we call **Test-Time Score Distillation (TSD)**.

Contributions.

1. **A reframing.** Training-free style transfer is a single *style-locality* axis whose two ends (Semantic Style Editing, Global Style Transfer) are both first-class tasks; we serve both with one operator and a single free knob (reverse-step count).
2. **An empirically tested regularization substitution.** We interleave bounded latent updates with the frozen reverse trajectory and show that an auxiliary fake-score network is unnecessary in the evaluated regime; adding one erases structured media rather than improving stability.
3. **A separable-axis evaluation protocol.** We evaluate style fidelity (CSD), subject fidelity (CLIP-T), and composition transfer (layout-leak) as three *separable* axes, complemented by a task-aware six-dimensional Agent audit, and show that a single style number rewards subject-destroying over-stylization — a failure only the joint reading exposes. (We also tested, and honestly reject, a

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generator-native single-forward style metric read from the spear; § 4.)

4. **A fair, strong baseline and a decisive result.** Against the same frozen model *directly instructed* to edit the image with its strongest style-transfer prompt, TSD reaches **2.5–2.7× its CSD** at matched subject fidelity across all 40 styles (with half the variance), and — unlike the native edit — can dissolve composition while keeping medium.

Related Work

Distillation and test-time score guidance. Distribution-matching distillation matches a student generator to a teacher through real-minus-fake score differences, with later variants improving the objective and separating a distribution-matching *shield* from a CFG-augmentation *spear* (Yin et al. 2024a,b; Liu et al. 2025). This decomposition connects to diffusion and score models' noise-conditional vector fields (Ho, Jain, and Abbeel 2020; Song et al. 2021): conditional contrasts yield classifier or classifier-free guidance, score distillation, and text-guided editing (Dhariwal and Nichol 2021; Ho and Salimans 2022; Poole et al. 2023; Hertz, Aberman, and Cohen-Or 2023; Nam et al. 2024; Chinchuthakun et al. 2025). Other test-time methods steer frozen samplers with external losses, perturbed attention, or annealed conditions (Bansal et al. 2023; Ahn et al. 2024; Hong 2024; Sadat et al. 2024). TSD instead specializes the spear to one test-time sample: it optimizes only the current latent and empirically substitutes the frozen reverse trajectory for a separately trained fake-score network.

Reference-conditioned style transfer and frozen-model editing. Classical style transfer matches feature statistics (Gatys, Ecker, and Bethge 2016; Huang and Belongie 2017); diffusion methods extend this idea through exemplar conditioning, learned style interfaces, attention sharing, or sampling-time transport (Yang et al. 2023; Sohn et al. 2023; Ye et al. 2023; Hertz et al. 2024; Chung, Hyun, and Heo 2024; Deng et al. 2024; Wang et al. 2024; Xu et al. 2025; He et al. 2025; Rout et al. 2025b). Related editing systems control cross-attention or reuse frozen diffusion features (Hertz et al. 2023; Tumanyan et al. 2023), while rectified-flow methods develop inversion and direct trajectory editing (Rout et al. 2025a; Kulikov et al. 2025). TSD belongs to the optimization-based, training-free-of-weights family, but its signal is the native same-state reference-conditioning contrast rather than replaced attention features, a trained adapter, or an explicit mask. Accordingly, we do not target exact motif replication through attention key/value manipulation; we target the transferable medium, palette, and texture statistics controlled by the style-locality axis.

Method

Figure 1 summarizes the operator derived in this section as an alternating cycle of task-direction distillation, a bounded latent update, and frozen-prior regularization.

Velocity Differences Are Score Differences

A rectified-flow transformer (Esser et al. 2024) predicts a velocity $v_\theta(z_t, t, c)$ at latent z_t, time t, conditioning c. For the flow-matching schedule (Lipman et al. 2023; Liu, Gong, and Liu 2023) the velocity is an affine function of the conditional score, $s_\theta(z_t, t, c) = -(z_t - (1-\sigma)\hat{x}_0)/\sigma^2$, so a *difference* of velocities under two conditionings equals, up to one positive schedule factor, a difference of conditional scores:

$$v_\theta(z, c_1) - v_\theta(z, c_2) \;\propto\; \nabla_z \log p_t(z \mid c_1) - \nabla_z \log p_t(z \mid c_2) \;=\; \nabla_z \log \frac{p_t(z \mid c_1)}{p_t(z \mid c_2)}. \quad (1)$$

Every velocity difference the frozen model gives us is therefore the gradient of a log-density ratio we may ascend (attract) or descend (repel). This is the whole toolbox: no auxiliary network, only frozen forward passes.

The Spear: CFG-Augmentation as the Style Driver

Decoupled distribution matching (Liu et al. 2025) writes its distillation gradient as

$$\nabla \;\propto\; \underbrace{(s_{\text{cond}} - s_{\text{fake}})}_{\text{shield (distribution match)}} \;+\; \underbrace{(\alpha - 1)\,(s_{\text{cond}} - s_{\text{uncond}})}_{\text{spear (CFG augmentation)}}. \quad (2)$$

The spear is a *conditional contrast* evaluated at the sample's own state, structurally the image-conditioned analogue of classifier-free guidance (Ho and Salimans 2022); it needs only the frozen teacher. We instantiate the style spear with two frozen velocities: v_{with}, conditioned on the reference image (the style evidence), and v_{without}, conditioned on the style *text* alone. The style guidance is

$$g \;=\; v_{\text{with}} - v_{\text{without}} \;\propto\; \nabla_z \log \frac{p_t(z \mid \text{reference})}{p_t(z \mid \text{style text})}, \quad (3)$$

the extra style evidence the reference *image* supplies beyond what words can.

The Regularization Substitution: The Frozen Reverse Process *Is* the Shield

The shield stabilizes repeated generator updates during distillation training. In DMD, the fake-score term discourages the learned generator from drifting away from the teacher distribution. Our setting differs: we optimize one latent while retaining the frozen model's ordinary T-step reverse trajectory. Each bounded TSD update is followed by a standard reverse step, which repeatedly reintroduces the frozen model prior. We therefore test an empirical regularization substitution: whether these inference dynamics make a separately learned fake-score regularizer unnecessary in the single-sample regime.

The trained shield is unnecessary in the evaluated regime. If the conditional contrast caused unstable drift, increasing its pre-normalization scale should expose artifacts. Instead, scales from 4 to 20 produce near-identical outputs because rms normalization removes that scale, and no collapse is observed across the tested trajectories. This does not establish formal distribution matching; it shows that an auxiliary fake-score network is not needed for the evaluated test-time procedure.

The trained shield is harmful in our implementation. Adding a real fake-score critic removes visible structured

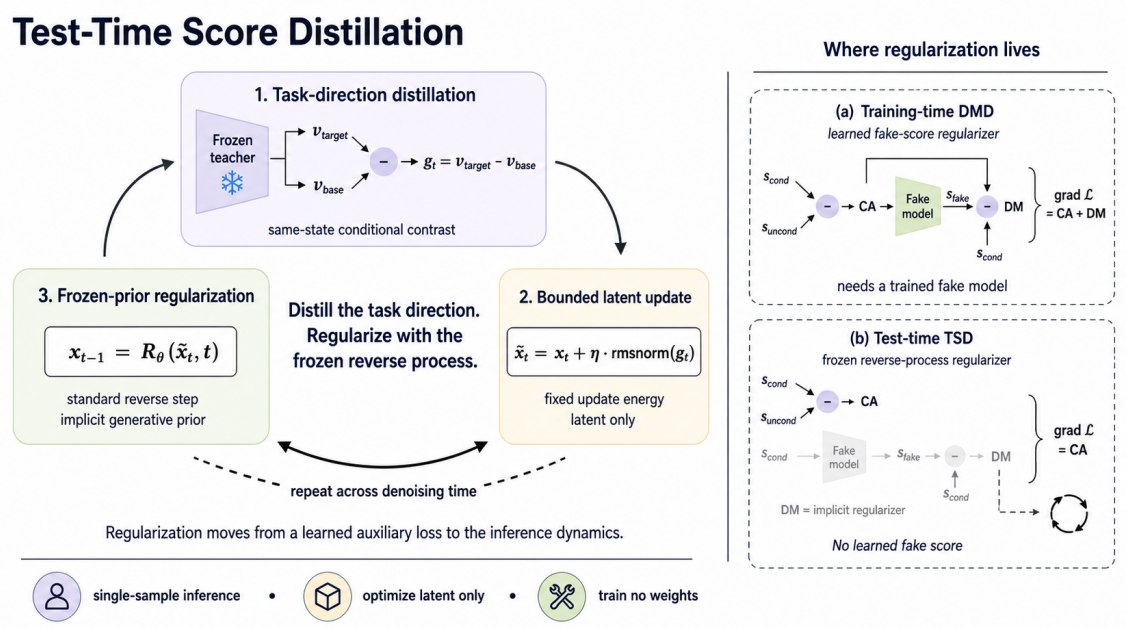


Figure 1: **Test-Time Score Distillation (TSD) alternates task-direction distillation with frozen-prior regularization.** *Left (the operator).* At each denoising state the frozen teacher's same-state conditional contrast $g_t = v_{\text{target}} - v_{\text{base}}$ supplies the task direction (a reference-conditioned minus a reference-free velocity); an rms-normalized pseudo-target bounds the latent update, $\tilde{x}_t = x_t + \eta\,\mathrm{rmsnorm}(g_t)$; the frozen model's standard reverse step R_θ then acts as an implicit generative-prior regularizer. *Right (where the regularization lives).* The decoupled distillation gradient splits into two terms: **CA** (*CFG-Augmentation*, the *spear* $s_{\text{cond}} - s_{\text{uncond}}$, where $s_{\text{cond}}/s_{\text{uncond}}$ are the reference-conditioned / reference-free scores) and **DM** (*Distribution Matching*, the *shield*, formed from a learned fake-score s_{fake}). **(a) is Decoupled DMD** (training-time): it realizes the shield with a *trained* fake-score network, so $\nabla\mathcal{L} = \text{CA} + \text{DM}$. **(b) is our TSD** (test-time): it drops the learned fake score and lets the frozen reverse process serve as the implicit regularizer, so $\nabla\mathcal{L} = \text{CA}$ alone — no auxiliary network is trained.

media: for example, mosaic tesserae revert toward a naturalistic rendering. Scheduled fake-score updates also fail to improve three styles with remaining headroom. These observations support using the frozen reverse process as the only regularizing dynamics, without claiming that the same conclusion holds for every model or sampler.

Dropping the shield and keeping the spear gives **Test-time Score Distillation (TSD)**.

The RMS-Normalized Pseudo-Target: a Fixed-Magnitude Extrapolation

We apply the spear as a pseudo-target in x_0-space rather than as a raw gradient. With $\hat{x}_0 = z - \sigma\, v_{\text{without}}$ the current denoised estimate, we form a detached target that is the estimate nudged along the *unit* style direction,

$$x_0^{\text{tgt}} = \mathrm{sg}\big(\hat{x}_0 + \eta\,\widehat{g}\big), \qquad \widehat{g} = \mathrm{rmsnorm}(g), \tag{4}$$

$$\mathcal{L} = \tfrac{1}{2}\big\|\hat{x}_0 - x_0^{\text{tgt}}\big\|^2, \tag{5}$$

so the latent update is exactly $\eta\,\widehat{g}$: a **fixed-magnitude step** of size η along the style direction, applied every inner iteration. Two consequences follow. First, because rms-normalization removes the overall scale of g, the guidance weight α and any linear coefficient are **washed out** — only the *direction* of g and the step size η affect the output. Second, a fixed-magnitude step is an *extrapolation past* the weak fixed point $v_{\text{with}} = v_{\text{without}}$, whereas minimizing $\|g\|^2$ would *interpolate to* that fixed point and saturate early; the former transfers structured texture the latter misses while preserving content.

Style Locality: One Operator, Two Modes, One Free Knob

Semantic Style Editing is the full spear on the reference. With $g = v_{\text{with}} - v_{\text{without}}$ attracting to the whole reference — its medium *and* its composition — the prompt's subject is rendered in the reference's medium *within the reference's composition*. Subject preservation is emergent: no mask, no segmentation, no region blending.

Global Style Transfer is the same spear with fewer reverse steps. Reducing the reverse-step count underdetermines the sample's structure, so the reference's specific composition never solidifies, while the per-step style spear still imprints the reference's medium and palette. Composition dissolves; medium remains. The step count is a *free, continuous* knob along the style-locality axis — from many steps (local, composition-preserving) to few steps (global, composition-dissolving).

Algorithm 1 TSD inference (one style transfer)

Require: frozen model, reference image, prompt; reverse steps T, inner iterations K, step size η

start from a noisy latent z_T
for reverse step $t = T$ **to** 1 **do**
  for $k = 1$ **to** K **do**
    read the model's velocity *with* the reference (v_{with}) and *without* it (v_{without})
    $\text{spear} \leftarrow$ direction of $(v_{\text{with}} - v_{\text{without}})$ // the style signal
    nudge z_t one small step η along the spear // change the latent only
  end for
  take one ordinary frozen reverse step to get z_{t-1} // the only regularizer
end for
return the image decoded from z_0

Why Velocity Space Stops at the Transferable Part (Analysis)

The frozen-velocity spear transfers the position-free part of style, not the exact motif layout — which is exactly what the axis should control. Reproducing a reference's *specific* tessellation or leading requires attention-level key/value manipulation, which we exclude. To characterize the boundary, we probe a taboo-clean, *position-only* construction: we compute the reference velocity with the reference tokens' positional identifiers scrambled, isolating the reference's appearance from its layout (this touches no attention statistics — only input positions consumed by the frozen model). Guided by this layout-free velocity, global transfer succeeds cleanly on *medium-dominant* styles (relief, halftone, stained-glass leading — the medium survives position scrambling) but collapses the subject on *figure-dominant painterly* styles (portrait mosaics, oil portraits), where medium and composition are entangled. The reverse-step knob is the robust instrument; the layout-free probe explains *why* certain references dissolve cleanly and others do not, and it supplies the generator-native metric of § 4.

Evaluation Protocol and Perceptual Diagnostics

We measure three separable axes, because style transfer is not one number. Style fidelity (is the medium transferred?), subject fidelity (is the prompt's subject present?), and composition transfer (is the reference's layout kept or dissolved?) can move independently, and a single metric hides failures. We report **CSD** (Contrastive Style Descriptor cosine similarity to the reference — style fidelity, designed to be content-invariant, though as we show below it is gameable at *extreme* over-stylization) (Somepalli et al. 2024), **CLIP-T** (image–text alignment to the content prompt — subject fidelity) (Radford et al. 2021), and a **layout-leak** measure (correlation of off-diagonal patch self-similarity — composition transfer). Layout-leak cleanly separates the two modes in our study: high for semantic editing (composition kept), low for global transfer (composition dissolved).

Table 1: Public-editor audit: primary-Agent means over five fixed styles (IDs 00, 09, 10, 23, 34) on the six 1–5 axes. Each public editor receives the raw reference, the same panda request, seed 65, and an explicit Semantic or Global instruction. This is a matched-input perceptual stress test, not a compute- or architecture-matched leaderboard.

Task	Method	Medium	Texture	Palette	Subject	Locality	Quality
Semantic	TSD	4.60	4.40	4.80	3.80	5.00	4.00
	Native MMDiT	2.40	2.40	2.80	5.00	1.80	4.40
	FireRed Edit 1.1	3.40	3.40	4.00	5.00	3.80	4.60
	Qwen Edit 2511	4.00	4.00	4.00	5.00	3.80	4.40
Global	TSD	4.80	4.80	5.00	4.20	3.40	4.40
	Native MMDiT	2.60	2.40	2.40	5.00	5.00	4.40
	FireRed Edit 1.1	2.00	2.20	2.60	5.00	3.00	4.00
	Qwen Edit 2511	4.00	4.20	4.20	4.20	2.00	4.00

CSD alone is misleading, which is why we pair it with CLIP-T. In the semantic grid, pushing the spear harder monotonically raises CSD but past a point *destroys the subject*: the highest-CSD configuration (CSD ≈ 0.82) has CLIP-T collapsed to ≈ 0.21 and layout-leak ≈ 0.82 — the panda is being over-painted into pure texture. The right operating point balances CSD and CLIP-T rather than maximizing CSD; reporting both makes over-stylization visible instead of rewarding it. This is a boundary artifact, not our operating regime: we verify in § 5 that our *benchmark* CSD advantage is not driven by it — the gap survives at matched layout-leak.

A task-aware Agent benchmark resolves the same trade-off perceptually. For each of 40 references, a primary VLM receives the reference and two anonymous, randomly ordered candidates for the matched task. It scores six independent 1–5 axes: medium/technique, texture/mark-making, palette/tone, subject fidelity, task locality, and visual quality. Locality means *composition preservation* for Semantic Style Editing and *composition independence* for Global Style Transfer. The primary judge covers all 80 paired task–style cells; a secondary model audits 68 cells, and failed API responses are not imputed. Prompts, raw responses, bootstrap intervals, and judge controls are provided in the supplement.

The Agent results localize the gain instead of hiding the cost. TSD is preferred in 39/40 Semantic and 37/40 Global pairs. Relative to the matched native edit, its mean gains on medium, texture, and palette are $+2.5$–$+2.9$ points, while subject fidelity is 0.6 points lower. Semantic composition preservation improves by $+3.45$ points. In Global transfer, however, composition independence is 1.63 points lower than the native new-composition edit: TSD transfers far more style but retains more reference layout, consistent with layout-leak. The two Agents agree on the preferred method in all 68 shared cells; their correlations are 0.96 on the three style axes and 0.93 on task locality.

Public-editor audit. On a fixed five-style audit against Qwen-Image-Edit-2511 (Wu et al. 2025) and FireRed-Image-Edit-1.1 (Super Intelligence Team 2026) (Table 1), TSD leads the reference-style axes: Semantic medium/texture/palette

average **4.60** versus 4.00 (Qwen), 3.60 (FireRed), and 2.53 (native); Global **4.87** versus 4.13, 2.27, and 2.47. The primary Agent selects TSD as the best-style candidate in $3/5$ Semantic and $5/5$ Global cells, and a full Gemini-3.5 Flash re-audit agrees on 80% of cells per task. The public editors retain stronger literal subject fidelity (5.00 versus TSD's 3.80 in Semantic) — the same style↔subject trade-off seen against the native edit — while the native path alone keeps best composition independence in Global. Per-axis cross-Agent correlations are in the supplement.

A generator-native style metric, explored and rejected (honest negative result). External style metrics such as CSD and DINO (Oquab et al. 2024) judge with a *separate* pretrained network; it is tempting to read style straight from the generator by re-evaluating the spear on the output, $\mathrm{SRSD}(x, R) = \|v_{\text{with}}(x; R) - v_{\text{without}}(x)\|_{\text{rms}}$, expecting a small residual when x already carries R's style. We tested this ($n = 20$, four configurations $\times$ five styles) and reject it. Its correlation with CSD is **positive, not negative** (Pearson $+0.79$, Spearman $+0.88$), and the correlation is carried almost entirely by over-stylized samples whose *content has collapsed* into a near-verbatim copy of the reference — exactly the samples CSD wrongly rewards. Controlling for spear strength, the residual correlation vanishes ($r = 0.20$, n.s.), and scale-correction does not rescue it. The spear magnitude marks *content collapse / composition copying*, not style faithfulness; flipping its sign would reward the pathology. This is the same style–composition entanglement that bounds velocity-space transfer (§ 3.6): a frozen-velocity scalar cannot isolate pure style. We therefore keep the three external, separable axes and report this as a cautionary null.

Experiments

Setup. A single frozen text-to-image rectified-flow transformer serves as both conditional and unconditional predictor (one model, shared runtime); all results are training-free of weights. Unless noted, generations use seed 65 and the prompt "a panda eating bamboo" on a 40-reference style set spanning mosaic, relief, oil, halftone pop-art, and stained-glass media. We report CSD, CLIP-T, layout-leak, and the task-aware Agent audit.

The Style-Locality Axis and the Parameter Study

The live knobs are step count / inner iterations, step size η, and per-token normalization; guidance weight is inert. Consistent with § 3.4, sweeping the guidance weight leaves the output unchanged (rms-normalization washes it out); we therefore grid the knobs that matter. For **Global Style Transfer** (few-step regime), CSD rises with step size, peaking at *num-steps 8,* $\eta = 0.12$ (CSD ≈ 0.63–0.64), above the default 8-step anchor (CSD 0.53), while layout-leak stays low (≈ 0.45, composition dissolved). For **Semantic Style Editing** (25-step regime), CSD rises with inner iterations and step size but over-stylizes at the top; the balanced optimum is *iters 5,* $\eta = 0.08$ (CSD 0.74, CLIP-T 0.28), versus the canonical anchor (CSD 0.70, CLIP-T 0.285). The complete grids for both normalization choices, including all CSD, CLIP-T, and layout-leak values, are reported in the supplement.

Table 2: Full benchmark at fixed seed 65 (40 styles, mean $\pm$ SD across styles). Higher CSD = more faithful medium; higher CLIP-T = subject present; layout-leak = composition transfer (high = kept, low = dissolved).

Method	CSD ↑	CLIP-T ↑	Layout-leak
TSD Semantic	**.719 ± .078**	.295 ± .030	.572 ± .146
TSD Global	**.615 ± .095**	.304 ± .026	.414 ± .108
Native keep-comp.	.267 ± .163	.319 ± .012	.317 ± .072
Native new-comp.	.247 ± .155	.315 ± .011	.299 ± .064

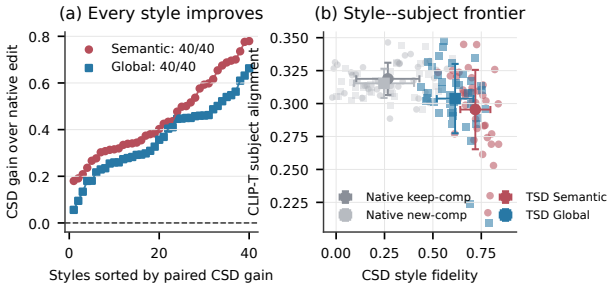


Figure 2: **TSD improves style fidelity across two locality regimes.** (a) All 40 paired CSD gains are positive. (b) CSD–CLIP-T frontier at fixed seed 65 ($n = 40$; mean $\pm$ cross-style SD). Qualitative outputs on two content prompts are in Figure 3.

Full Benchmark Against the Frozen Model's Native Edit

Baseline design (a fair, strong baseline, not a straw man). The most honest baseline is the *same* frozen model *directly instructed* to edit the reference with text — its native image+text edit. We first verify that this is a genuinely different mechanism: the frozen reverse process in our pipeline does not itself consume the edit image (the reference enters only through the spear), so "our operator with zero spear" is prompt-only text-to-image, not a native edit. We therefore run the model's own native edit pipeline at its documented best settings. Because that checkpoint exposes no system-prompt channel, we fold a strong expert style-transfer instruction into the prompt (and a negative prompt that deliberately does *not* penalize flat media, to avoid disadvantaging pop-art and stained-glass). This strong prompting substantially improves the baseline (mean CSD from $0.24 \rightarrow 0.36$ for the composition-preserving instruction, $0.03 \rightarrow 0.29$ for the new-composition instruction), so the comparison is against the baseline's best effort.

Result (40 styles). Even at its best, the native edit is dominated on style fidelity by a wide margin across references (Table 2). TSD reaches CSD **0.72** $\pm$ **0.08** (semantic) and **0.61** $\pm$ **0.09** (global) versus **0.27** $\pm$ **0.16** and **0.25** $\pm$ **0.16** for the native edit — a **2.5–2.7**$\times$ **style-fidelity gain** (2.69$\times$ semantic, 2.48$\times$ global). The most robust form of the claim is not the mean: **ours wins CSD on every one of the 40 styles, in both pairings (40/40 semantic, 40/40**



Figure 3: **The medium transfers across content, not just one subject.** Five media-diverse styles (rows) are each applied to two very different content prompts — a panda and a lighthouse. The leftmost column is the shared style reference; within each content block the columns are the two matched native edits (keep-/new-composition) and TSD Semantic / TSD Global. TSD renders *both* subjects in the reference medium, whereas the native image+text edits default to a generic realistic rendering that ignores the medium. The panda is the quantified benchmark subject; the lighthouse is an *unseen* qualitative probe. Seeds are selected per cell from $\{65, 202, 777\}$ for legibility; all quantitative results use fixed seed 65. Four further content prompts, each with paired quantitative wins, are in the supplement.

global) — despite the native edit's large cross-style dispersion (whose lower tail overlaps ours), it never wins a single style (Figure 2a). A spot-check on the native edit's worst style (a neon-anime reference, its CSD $\approx$ 0) shows why: the native edit defaults to a generic realistic panda that ignores the reference medium entirely, so its near-zero CSD is a genuine style-transfer failure, not a scoring artifact. Two things sharpen the result. First, the gap is at *matched or slightly lower* subject fidelity: the native edit keeps CLIP-T marginally higher (≈ 0.315 vs. 0.30) because it stylizes less — an honest style↔subject trade-off, visible on the CSD–CLIP-T frontier (Figure 2b). Second, the native edit's CSD dispersion across references is *twice* ours (SD 0.16 vs. 0.08): it is not only weaker but far less consistent across styles, succeeding on genuinely relief-like media (*e.g.*, bronze) and reverting to a *generic* engraving/line-art look elsewhere, missing the specific tesserae/impasto/halftone/leading that TSD transfers consistently (Figure 3). Finally, when instructed to change composition, the native edit *loses the medium* (its new-composition variant scores CSD 0.25, below its own composition-preserving 0.27), whereas TSD's step-knob dissolves composition while *keeping* the medium (global CSD 0.61) — the native edit cannot separate style from layout, which is exactly the global-transfer end of the axis.

Is the style-fidelity gap real style, or just more reference-copying? Because CSD can be gamed by near-verbatim reference copies at extreme over-stylization (§ 4), and our outputs carry higher layout-leak than the native edit, a skeptic could ascribe the CSD gap to copying rather than style. It is not. We lead the pure-style claim with *Global Style Transfer*, whose whole point is to *dissolve* the reference's composition: its layout-leak (0.41) is close to the native edit's (0.30), yet it still wins CSD 0.61 vs. 0.25 — a 2.5× gap at comparable composition-copying. More directly, we restrict to the 14 styles where our global output copies *no more* layout than the native edit (layout-leak $\leq$ its mean$+1\sigma = 0.36$) and find ours still wins CSD **14/14** (0.61 vs. 0.24); within the global grid the CSD–leak correlation is weak ($+0.29$). The gap survives at matched layout-leak, so it is style fidelity, not composition-copying.

CLIP-T is computed against the literal content prompt ("a panda eating bamboo"), uniformly across all four methods,

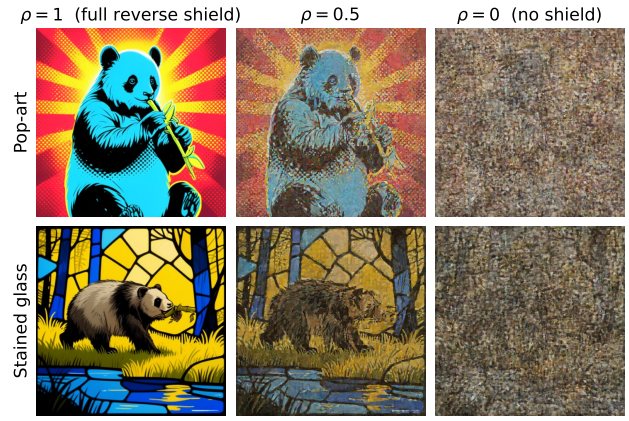


Figure 4: **The frozen reverse step is the regularizer.** Scaling the frozen scheduler displacement $z \leftarrow z + \rho\,(\mathrm{step}(z) - z)$ by $\rho \in \{1, 0.5, 0\}$ at a fixed task update (two representative styles). At $\rho = 1$ (the original reverse process) the stylization is clean; at $\rho = 0$ the scheduler is still evaluated but its displacement is withheld, so the fixed-magnitude task updates alone drive the latent off-distribution into noise; $\rho = 0.5$ interpolates. The collapse is style-independent (all five styles in the supplement).

so the comparison is fair; scoring against the fuller style-template prompt shifts every method up by a near-constant ≈ 0.03 and preserves the ordering. CSD and layout-leak are prompt-independent.

The CSD advantage is robust across content prompts. To confirm the win is not an artifact of the single benchmark prompt, we re-ran a media-diverse 5-style subset (mosaic, bronze, oil, pop-art, stained-glass) across four additional content prompts (a seaside castle, a cat on a windowsill, a sunflower field, an aged fisherman). Ours wins CSD on every style×prompt cell — **20/20 in both pairings, zero flips** — with per-prompt ratios of 1.55–2.31× (semantic) and 1.65–2.20× (global). The multiplier on this subset is $\approx$ 1.9–2×, not the 2.5–2.7× headline: the headline is the full 40-style figure, whereas this deliberately media-diverse subset is harder and its own single-prompt anchor already sits at $\approx 2\times$. The style↔subject trade-off is prompt-dependent — near-matched CLIP-T on subject-light prompts, but on the *portrait* prompt our CLIP-T falls further (gap up to ≈ 0.08) as the spear over-stylizes the face — the same honest trade-off, more visible where subject identity is strong.

Ablations

The frozen reverse step is load-bearing. We scale the scheduler displacement $z \leftarrow z + \rho\,(\mathrm{step}(z) - z)$ by $\rho \in \{1, 0.5, 0\}$ at a fixed task update (Figure 4). At $\rho = 0$ the model is still evaluated but its displacement withheld, so the task updates alone drive the latent and every style drifts off-distribution into noise; $\rho = 0.5$ partially degrades. This is not merely disabling the sampler — the task optimization is unchanged — so it is the reverse process, not the task update, that keeps the optimized latent coherent. **A trained shield is unnecessary and harmful.** Conversely, adding a trained fake-score shield (the literal DMD term) degrades style: structured media are pulled back toward naturalism (mosaic tesserae lost). The frozen reverse process alone is the better regularizer in the training-free regime. **Spear direction vs. magnitude.** The fixed-magnitude rms-normalized step (extrapolation) transfers texture that magnitude-minimizing objectives (interpolation to the fixed point) miss, at equal content preservation. **Guidance weight.** Sweeping α leaves outputs unchanged, confirming rms-normalization removes scale — the direction is the signal.

Evaluation Robustness and Failure Modes

The Agent audit is judge-robust. To reduce dependence on a single model family, a third VLM (a local Gemma-4-family checkpoint) re-scored the 95 calibration outputs. It agrees moderately with both Gemini judges (Pearson 0.59–0.69; at least 82% of ratings fall within one point) and, decisively, passes the pre-specified wrong-reference control: its mean style score drops from 4.21 for the correct reference to 1.33 for a reproducibly assigned wrong reference, and it scores the correct reference higher on 89.5% of outputs. The audit therefore reads reference-specific style rather than assigning uniformly high scores; it remains a judge-robustness test, not a human study.

Where TSD gives up subject alignment. The trade-off concentrates on subject-heavy references, as the portrait case above shows; on already-graphic references the medium is instead near-saturated, so the gain is neutral rather than negative. In Global transfer the medium survives while composition dissolves, yet the reference layout is retained more than in the native new-composition edit — the honest cost of transferring more style, visible as the higher layout-leak in Table 2. None of these is a scoring artifact; each is the same style–subject trade-off the frontier is built to expose, which is why we report paired axes rather than a single style score.

Cost. TSD is training-free of weights but optimization-based at inference: the reported settings run five inner latent updates per reverse step (Semantic) or a few-step schedule (Global), adding compute over a plain sample while training nothing. Memory is dominated by the frozen backbone, so a single generation fits in about 24 GB and completes in roughly a minute on one GPU; we optimize only the current latent and keep no auxiliary network.

Conclusion

Training-free style transfer on a frozen transformer is a single *style-locality* axis served by one operator: the CFG-augmentation spear of decoupled distribution matching, applied as a fixed-magnitude rms-normalized latent step. Its non-obvious ingredient is an empirical *regularization substitution*: interleaving bounded updates with the frozen reverse trajectory removes any need for a learned fake-score network in the evaluated regime. Across 40 styles this lifts mean CSD by 2.48–2.69× over the same model's strongest native edit, at a modest subject-alignment cost — with no training, no auxiliary fake-score network, and no mask or segmentation.

References

Ahn, D.; Cho, H.; Min, J.; Jang, W.; Kim, J.; Kim, S.; Park, H. H.; Jin, K. H.; and Kim, S. 2024. Self-Rectifying Diffusion Sampling with Perturbed-Attention Guidance. In *ECCV*.

Bansal, A.; Chu, H.-M.; Schwarzschild, A.; Sengupta, S.; Goldblum, M.; Geiping, J.; and Goldstein, T. 2023. Universal Guidance for Diffusion Models. In *CVPR Workshops*.

Chinchuthakun, W.; Saengja, T.; Tritrong, N.; Rewatbowornwong, P.; Khungurn, P.; and Suwajanakorn, S. 2025. LUSD: Localized Update Score Distillation for Text-Guided Image Editing. In *ICCV*.

Chung, J.; Hyun, S.; and Heo, J.-P. 2024. Style Injection in Diffusion: A Training-Free Approach for Adapting Large-Scale Diffusion Models for Style Transfer. In *CVPR*.

Deng, Y.; He, X.; Tang, F.; and Dong, W. 2024. Z*: Zero-Shot Style Transfer via Attention Reweighting. In *CVPR*.

Dhariwal, P.; and Nichol, A. 2021. Diffusion Models Beat GANs on Image Synthesis. In *NeurIPS*.

Esser, P.; Kulal, S.; Blattmann, A.; Entezari, R.; Müller, J.; Saini, H.; Levi, Y.; Lorenz, D.; Sauer, A.; Boesel, F.; Podell, D.; Dockhorn, T.; English, Z.; Lacey, K.; Goodwin, A.; Marek, Y.; and Rombach, R. 2024. Scaling Rectified Flow Transformers for High-Resolution Image Synthesis. In *ICML*.

Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2016. Image Style Transfer Using Convolutional Neural Networks. In *CVPR*.

He, H.; Hu, M.; Zheng, C.; Wang, C.; and Cham, T.-J. 2025. Semantix: An Energy-Guided Sampler for Semantic Style Transfer. In *ICLR*.

Hertz, A.; Aberman, K.; and Cohen-Or, D. 2023. Delta Denoising Score. In *ICCV*.

Hertz, A.; Mokady, R.; Tenenbaum, J.; Aberman, K.; Pritch, Y.; and Cohen-Or, D. 2023. Prompt-to-Prompt Image Editing with Cross Attention Control. In *ICLR*.

Hertz, A.; Voynov, A.; Fruchter, S.; and Cohen-Or, D. 2024. Style Aligned Image Generation via Shared Attention. *ACM Transactions on Graphics*, 43(4).

Ho, J.; Jain, A.; and Abbeel, P. 2020. Denoising Diffusion Probabilistic Models. In *NeurIPS*.

Ho, J.; and Salimans, T. 2022. Classifier-Free Diffusion Guidance. In *NeurIPS Workshop on Deep Generative Models and Downstream Applications*.

Hong, S. 2024. Smoothed Energy Guidance: Guiding Diffusion Models with Reduced Energy Curvature of Attention. In *NeurIPS*.

Huang, X.; and Belongie, S. 2017. Arbitrary Style Transfer in Real-Time with Adaptive Instance Normalization. In *ICCV*.

Kulikov, V.; Kleiner, M.; Huberman-Spiegelglas, I.; and Michaeli, T. 2025. FlowEdit: Inversion-Free Text-Based Editing Using Pre-Trained Flow Models. In *ICCV*.

Lipman, Y.; Chen, R. T. Q.; Ben-Hamu, H.; Nickel, M.; and Le, M. 2023. Flow Matching for Generative Modeling. In *ICLR*.

Liu, D.; Gao, P.; Liu, D.; Du, R.; Li, Z.; Wu, Q.; Jin, X.; Cao, S.; Zhang, S.; Li, H.; and Hoi, S. 2025. Decoupled DMD: CFG Augmentation as the Spear, Distribution Matching as the Shield. *arXiv preprint arXiv:2511.22677*.

Liu, X.; Gong, C.; and Liu, Q. 2023. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. In *ICLR*.

Nam, H.; Kwon, G.; Park, G. Y.; and Ye, J. C. 2024. Contrastive Denoising Score for Text-Guided Latent Diffusion Image Editing. *arXiv preprint arXiv:2311.18608*.

Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H.; Szafraniec, M.; Khalidov, V.; Fernandez, P.; Haziza, D.; Massa, F.; El-Nouby, A.; Assran, M.; Ballas, N.; Galuba, W.; Howes, R.; Huang, P.-Y.; Li, S.-W.; Misra, I.; Rabbat, M.; Sharma, V.; Synnaeve, G.; Xu, H.; Jégou, H.; Mairal, J.; Labatut, P.; Joulin, A.; and Bojanowski, P. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*.

Poole, B.; Jain, A.; Barron, J. T.; and Mildenhall, B. 2023. DreamFusion: Text-to-3D using 2D Diffusion. In *ICLR*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*.

Rout, L.; Chen, Y.; Ruiz, N.; Caramanis, C.; Shakkottai, S.; and Chu, W.-S. 2025a. Semantic Image Inversion and Editing using Rectified Stochastic Differential Equations. In *ICLR*.

Rout, L.; Chen, Y.; Ruiz, N.; Kumar, A.; Caramanis, C.; Shakkottai, S.; and Chu, W.-S. 2025b. RB-Modulation: Training-Free Personalization of Diffusion Models using Stochastic Optimal Control. In *ICLR*.

Sadat, S.; Buhmann, J.; Bradley, D.; Hilliges, O.; and Weber, R. M. 2024. CADS: Unleashing the Diversity of Diffusion Models through Condition-Annealed Sampling. In *ICLR*.

Sohn, K.; Ruiz, N.; Lee, K.; Chin, D. C.; Blok, I.; Chang, H.; Barber, J.; Jiang, L.; Entis, G.; Li, Y.; Hao, Y.; Essa, I.; Rubinstein, M.; and Krishnan, D. 2023. StyleDrop: Text-to-Image Generation in Any Style. In *NeurIPS*.

Somepalli, G.; Gupta, A.; Gupta, K.; Palta, S.; Goldblum, M.; Geiping, J.; Shrivastava, A.; and Goldstein, T. 2024. Measuring Style Similarity in Diffusion Models. *arXiv preprint arXiv:2404.01292*.

Song, Y.; Sohl-Dickstein, J.; Kingma, D. P.; Kumar, A.; Ermon, S.; and Poole, B. 2021. Score-Based Generative Modeling through Stochastic Differential Equations. In *ICLR*.

Super Intelligence Team. 2026. FireRed-Image-Edit-1.0 Technical Report. *arXiv preprint arXiv:2602.13344*.

Tumanyan, N.; Geyer, M.; Bagon, S.; and Dekel, T. 2023. Plug-and-Play Diffusion Features for Text-Driven Image-to-Image Translation. In *CVPR*.

Wang, H.; Spinelli, M.; Wang, Q.; Bai, X.; Qin, Z.; and

Chen, A. 2024. InstantStyle: Free Lunch towards Style-Preserving in Text-to-Image Generation. *arXiv preprint arXiv:2404.02733*.

Wu, C.; Li, J.; Zhou, J.; et al. 2025. Qwen-Image Technical Report. *arXiv preprint arXiv:2508.02324*.

Xu, R.; Xi, W.; Wang, X.; Mao, Y.; and Cheng, Z. 2025. StyleSSP: Sampling StartPoint Enhancement for Training-Free Diffusion-Based Method for Style Transfer. In *CVPR*.

Yang, B.; Gu, S.; Zhang, B.; Zhang, T.; Chen, X.; Sun, X.; Chen, D.; and Wen, F. 2023. Paint by Example: Exemplar-Based Image Editing with Diffusion Models. In *CVPR*.

Ye, H.; Zhang, J.; Liu, S.; Han, X.; and Yang, W. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. *arXiv preprint arXiv:2308.06721*.

Yin, T.; Gharbi, M.; Park, T.; Zhang, R.; Shechtman, E.; Durand, F.; and Freeman, W. T. 2024a. One-Step Diffusion with Distribution Matching Distillation. In *CVPR*.

Yin, T.; Gharbi, M.; Zhang, R.; Shechtman, E.; Durand, F.; Freeman, W. T.; and Park, T. 2024b. Improved Distribution Matching Distillation for Fast Image Synthesis. *arXiv preprint arXiv:2405.14867*.